Optimization for Deep Learning (OPT4DL)

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Stochastic Gradient Descent (SGD) without the G(radient)

Scenarios with no gradient



- Recall, we typically aim to solve

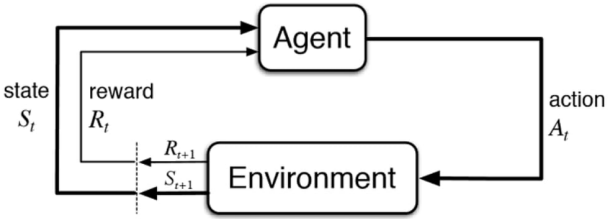
$$\min_w \frac{1}{n} \sum_{i=1}^{n} \ell(w, z_i) \tag{1}$$

- A natural stochastic gradient is to sample a mini batch of samples $z_t^1, \ldots, z_t^B$ and compute

$$g_t = \frac{1}{B} \sum_{j=1}^{B} \nabla \ell(w_t, z_t^j) \tag{2}$$

- However, we may not be able to, or want to, do this in many scenarios
  - (Online) Reinforcement Learning
  - Adversarial attacks
  - Fine-tuning of Large Language Models (LLMs)

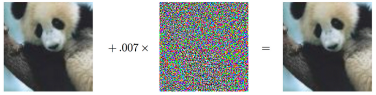
Example: (Online) Reinforcement Learning

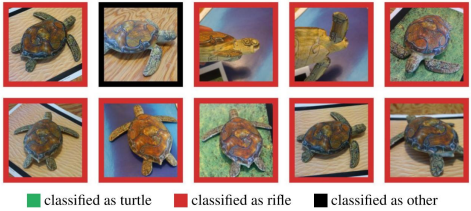


- Agent only gets to see the scalar reward, i.e., the reward for the played action.
- In certain RL models, the task can be reduced to an online optimization problem with linear utility functions $f_t(w) = \langle w, r_t \rangle$ where w is our policy and r_t is the reward vector
- The gradient is r_t but the agent only gets to see $R_t = r_t(i_s)$ where action a_{i_s} is the played action

Example: Adversarial Attacks

- Goal: Given a classifier and a small dataset, make the DL classily images incorrectly





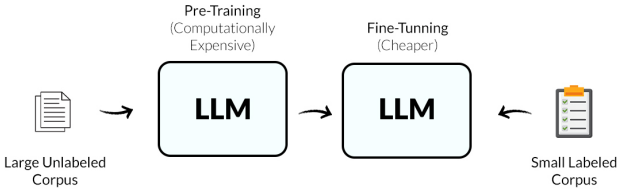
- We only know the images and the returned predictions. Let y_i be the label of image x_i

$$\min_{\delta} \ \text{score}_{w^*}(y_i|c_i+\delta) + \lambda\|\delta\| \tag{3}$$

  for some perturbation parameter δ.

- We do not know the weights of the DL model w^*, hence we cannot use gradient-based methods like SGD that need the gradient $\nabla \text{score}_{w^*}(y_i|c_i+\delta)$.

LLM fine-tuning



- As LLMs grow in size, the substantial memory overhead from gradient computation presents a significant challenge.
- Crucial to address, especially for applications like on-device training where memory efficiency is paramount.

SGD without the G: Intuition



- Let us recall the definition of derivatives for scalar functions

$$\begin{aligned} f'(w) &= \lim_{\mu \to 0} \frac{f(w+\mu) - f(w)}{\mu} \\ &= \lim_{\mu \to 0} \frac{f(w) - f(w-\mu)}{\mu} \\ &= \lim_{\mu \to 0} \frac{f(w+\mu) - f(w-\mu)}{2\mu} \end{aligned} \tag{4}$$

- Let $\mu > 0$ and fixed, then we expect

$$g^{\mu}(w) = \frac{f(w+\mu) - f(w-\mu)}{2\mu} \approx f'(w) \tag{5}$$

- We only need two function evaluations and no gradient calculation
- We expect to incur some bias depending on μ which vanishes as $\mu \to 0$

SGD without the G: Intuition (Cont'd)



- We can view g^{μ} as an expectation over random directions in $\mathbb{R}$ (i.e., left and right)
  - Let $u \sim \mathcal{U}\{-1, 1\}$, i.e., a vector in $\mathbb{R}$ lying on the surface of unit sphere.
  - Consider
$$g^{\mu u}(w) = \frac{f(w + \mu u) - f(w)}{\mu} u \tag{6}$$
    to be a random variable
- Then,
$$\begin{aligned} \mathbb{E}_u[g^{\mu u}(w)] &= \frac{1}{2}\frac{f(w+\mu) - f(w)}{\mu} - \frac{1}{2}\frac{f(w-\mu u) - f(w)}{\mu} u \\ &= \frac{f(w+\mu) - f(w-\mu)}{2\mu} = g^{\mu}(w) \approx f'(w) \end{aligned} \tag{7}$$
  when μ is small.
- Thus, $g^{\mu u}(w)$ is an unbiased estimator of $g^{\mu}(w)$, an approximation of the derivative of our loss function
- How to go to $\mathbb{R}^d$?

Zero-Order Gradient Estimator



- Recall: $u \sim \mathcal{U}\{-1, 1\}$, i.e., a vector in $\mathbb{R}$ lying on the surface of unit sphere.
- In general, we just sample u from the surface of unit sphere in $\mathbb{R}^d$
- Our estimator then becomes

$$g^{\mu u}(w) = \frac{f(w + \mu u) - f(w)}{\mu} u, \qquad u \sim \mathbb{S}_d \tag{8}$$

- Called zero-order estimator as it uses zeroth order information, i.e., function values only as opposed to first order (Gradient) or second order (Hessian) information.
- We can also sample u from a Gaussian distribution $\mathcal{N}(0, I_d)$ (e.g., for simpler mathematical analysis) given that

$$u \sim \mathcal{N}(0, I_d) \Leftrightarrow \frac{u}{\|u\|} \sim \mathbb{S}_d \tag{9}$$

ZO-SGD Method

- Consider $\min_w f(w) := \mathbb{E}_z[\ell(w, z)]$
- Initialize w_1,learning rate η, smoothing rate μ
- For $t = 1, \ldots, T$ do
  - sample a random direction $u_t \sim \mathcal{N}(0, I_d)$ or $u \sim \mathbb{S}_d$
  - sample a mini batch $z_t^1, \ldots, z_t^B$
  - form the update vector

$$g_t^{\mu u} = \frac{\frac{1}{B}\sum_{j=1}^{B} \ell(w_t + \mu u_t, z_t^j) - \ell(w_t, z_t^j)}{\mu} u_t \tag{10}$$

  - update the parameter $w_{t+1} = w_t - \eta g_t^{\mu u}$
- To lower the variance, we can sample multiple (a mini batch of) random directions as well

$$g_t^{\mu u} = \frac{1}{K}\sum_{k=1}^{K} \frac{\frac{1}{B}\sum_{j=1}^{B} \ell(w_t + \mu u_t^k, z_t^j) - \ell(w_t, z_t^j)}{\mu} u_t^k, \qquad u_t^k \sim \mathcal{N}(0, I_d) \tag{11}$$

ZO Estimation as Smoothing



- Recall $g^{\mu u}(w) = \frac{f(w+\mu u)-f(w)}{\mu} u$ and that

$$\mathbb{E}_u[g^{\mu u}(w)] = \frac{f(w+\mu) - f(w-\mu)}{2\mu} = g^{\mu}(w) \tag{12}$$

- So is $g^{\mu}(w)$ the true gradient for any function?
- In turns out

$$g^{\mu}(w) = \nabla f^{\mu}(w), \qquad f^{\mu} := \mathbb{E}_u[f(w+\mu u)] \tag{13}$$

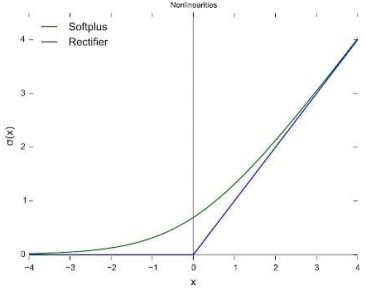
  and f^{μ} is called the smoothed version of f such that when $u \sim \mathcal{N}(0, I_d)$ the technique is called Gaussian Smoothing
- Called smoothed since its sensitivity properties are nicer than the original function f
  - f^{μ} has both lower Lipschitzness and Smoothness parameters
  - f^{μ} is smooth even if f is not! $L(f^{\mu}) \propto 1/\mu$
- μ controls the smoothness-approximation tradeoff

ZO Estimation as Smoothing (Cont'd)

- Recall

$$g^{\mu}(w) = \nabla f^{\mu}(w), \qquad f^{\mu} := \mathbb{E}_u[f(w + \mu u)] \tag{14}$$

- Intuition: Expectation is an integral and integral tends to make the functions smoother/nicer
- Indeed, f^{μ} is nothing but the convolution of f and the PDF of u. Recall: $f * g = \int f(\tau)g(w - \tau) \cdot d\tau$
- This idea is also useful when analyzing DL models with non-smooth activations, e.g., ReLU.



ZO-SGD for f as SGD for f^μ



- Recall $g^\mu(w) = \nabla f^\mu(w)$ and we used an unbiased estimator of $g^\mu(w)$ to minimize f without gradient calculation (ZO-SGD)
- Thus, ZO-SGD is effectively SGD for the following problem

$$\min_w f^\mu(w) := \mathbb{E}_u[f(w + \mu u)] = \mathbb{E}_{z,u}[\ell(w + \mu, z)] \tag{15}$$

- As μ tends to zero the two problem become equivalent
- This idea can be used to prove

Theorem

If $\mu \propto \frac{1}{d\sqrt{T}}$ and $\eta \propto \frac{1}{\sqrt{dT}}$ the iterates of ZO-SGD satisfy

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}_{z_{\mathbf{1}:T}, u_{\mathbf{1}:T}} \|\nabla f(w_t)\|^2 = \mathcal{O}\Big(\frac{d}{T} + \sqrt{\frac{d}{T}}\Big). \tag{16}$$